

Subspace Constraints

$$\begin{cases} \min: f(x) \\ \text{st: } x \in V, \end{cases} \quad V \text{ is a subspace}$$

→ $\min: \underbrace{f(x)}_f + \underbrace{I_V(x)}_g$

$$\text{DR: } x_{k+1} = \text{prox}_{\lambda f}(y_k)$$

$$y_{k+1} = y_k + \text{Proj}_V(2x_{k+1} - y_k) - x_{k+1}$$

(check) :

$$\begin{aligned} x_{k+1} &= \text{prox}_{\lambda f}(x_k - z_k) \\ z_{k+1} &= \text{Proj}_{V^\perp}(z_k + 2x_{k+1} - x_k) \end{aligned}$$

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Applying to Separable Problems

$$\min: f_1(x) + f_2(Ax), \quad f_1, f_2 \text{ have easy prox operators}$$

$$\begin{aligned} \min: & f_1(x_1) + f_2(x_2) \\ \text{st: } & (x_1, x_2) \in V : \boxed{x_2 = Ax_1} \end{aligned}$$

$$\text{DR: } f(x_1, x_2) = f_1(x_1) + f_2(x_2)$$

$$g = \mathbb{I}_V$$

Check: $\text{prox}_{\lambda f}$ is separable

Check: Proj_V is a linear operator

Explicitly Separable Functions

$$\min: \sum_{j=1}^n f_j(x_j) + \sum_{i=1}^m g_i(A_{i1}x_1 + \dots + A_{in}x_n)$$

$$\max: - \sum_{j=1}^n f_j^*(-A_{1j}^T z_1 - \dots - A_{mj}^T z_m) - \sum_{i=1}^m g_i^*(z_i)$$

If f_j^* smooth, can use dual prox grad

$$\hat{x}_j = \arg\min_{x_j} f_j(x_j) + \sum_i z_i^T A_{ij} x_j$$

$$z_i^+ = \text{prox}_{\lambda g_i^*}(z_i + \lambda \sum_j A_{ij}^T \hat{x}_j)$$

Check.

DR: $\min: \sum f_j(x_j) + \sum g_i(y_{i1} + \dots + y_{in})$

st: $y_{ij} = A_{ij} x_j$

$f = \sum f_j + \sum g_i$

$\text{prox}_f = ?$

show it decomposes
into ind. prox
for f_j & g_i .

Adding Splitting Variables

Compare the above to:

$$\min: \sum f_j(x_j) + \sum g_i(y_{i1} + \dots + y_{in})$$

$$\text{st: } x_{ij} = x_j //$$

$$y_{ij} = A_{ij} x_{ij}$$

Ex: write updates explicitly

Application: Total Variation Denoising/Deblurring

b is a blurred + noisy image,

$$\min: \|Kx - b\|_1 + \gamma \|Dx\|_{TV}$$

$$\text{st: } \underline{0 \leq x \leq 1}$$

Apply DR : $f_1(x) = \mathbb{I}_{[0,1]^n}(x)$

$$f_2(u, v) = \|u - b\|_1 + \gamma \|v\|_{TV}$$

$$\min: f_1(x) + f_2(Ax), \quad A = \begin{bmatrix} K \\ D \end{bmatrix}$$

$$Ax = \begin{pmatrix} Kx \\ Dx \end{pmatrix} \\ u, v$$

Exercise : Find explicit updates for DR

Optional : show that solving the linear system that arises, can be done faster by FFT.